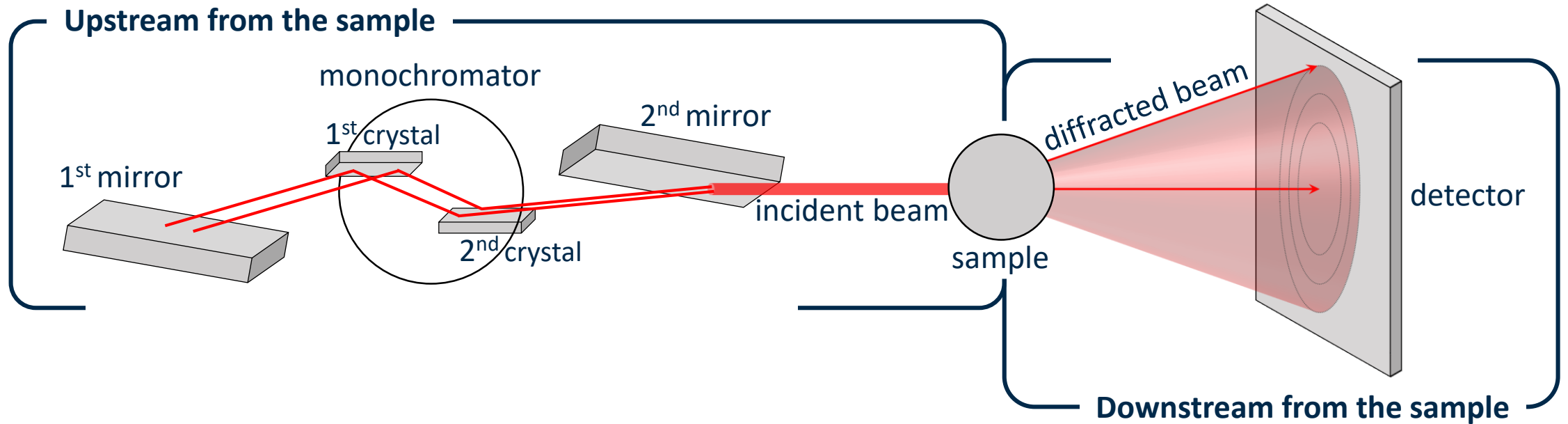


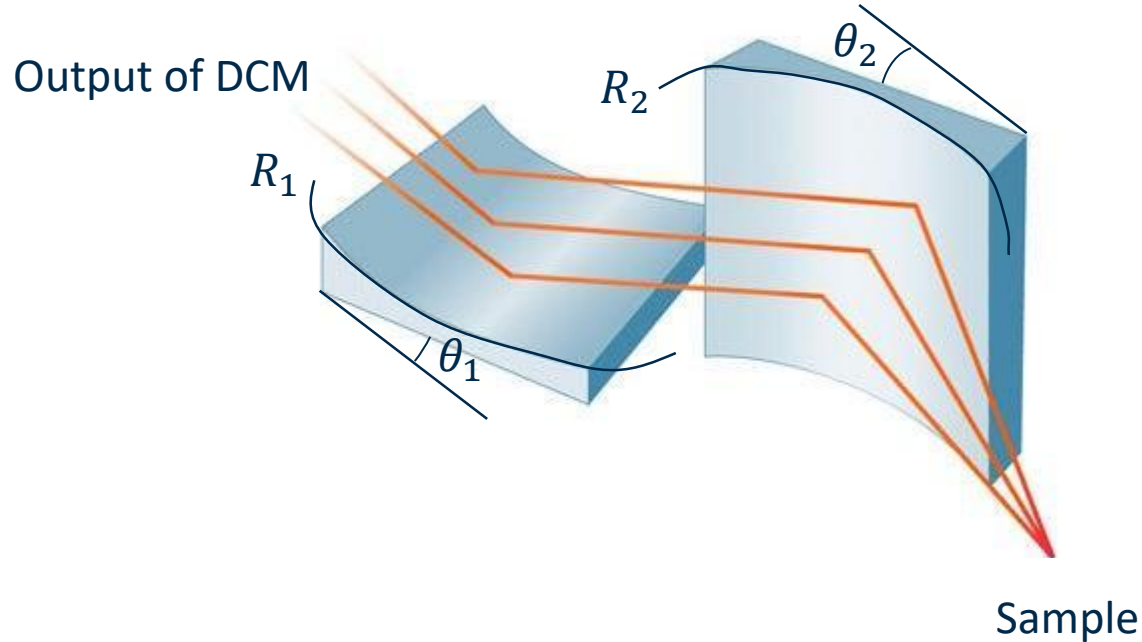
Use Case: Beamline alignment



- Choose **Beam Shape** (+ maximize intensity)
- Often **Iterative Process** with a lot a degrees of freedom → **takes a lot of time!!**

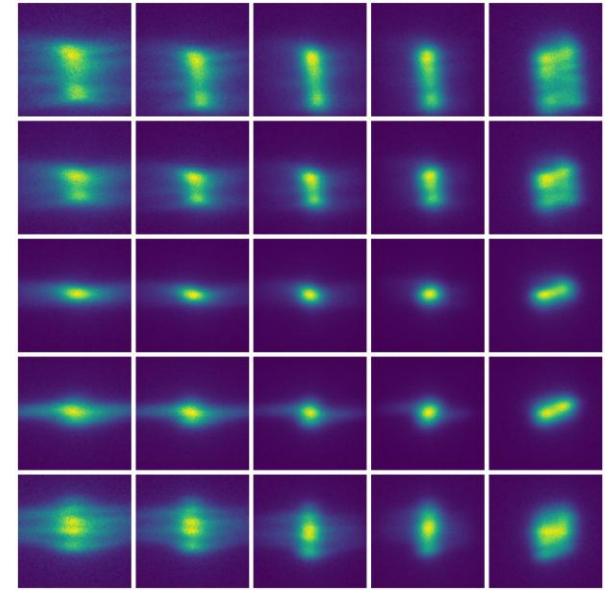
Simplified Use Case: KB mirrors

- A set of Kirkpatrick-Baez mirrors

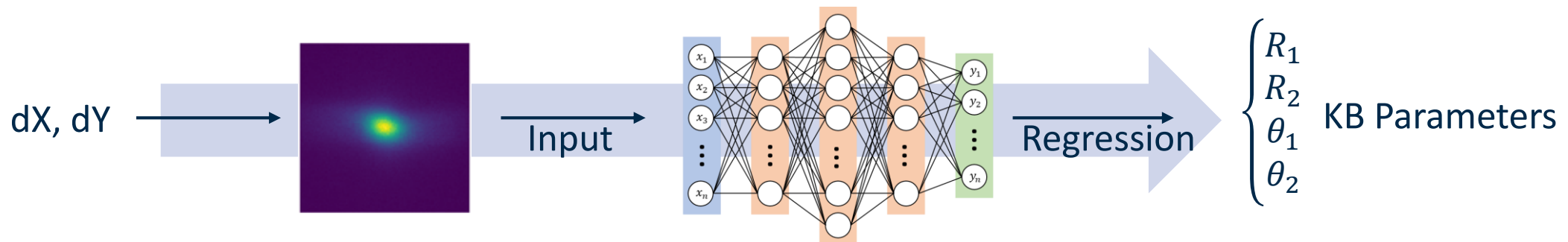


- Parameters: $\begin{cases} R_1 \\ R_2 \\ \theta_1 \\ \theta_2 \end{cases} \rightarrow$

Wide Range of different Beams



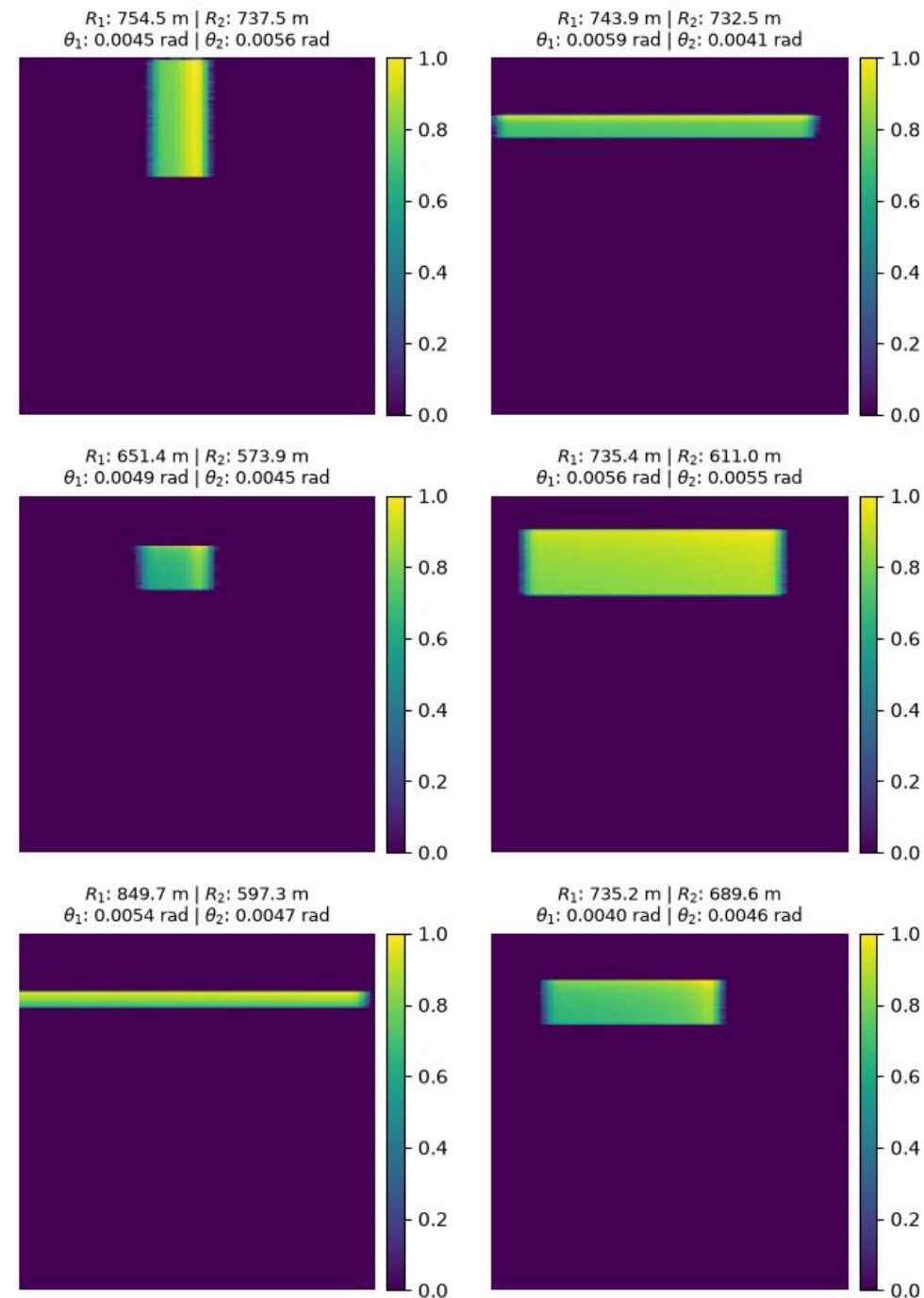
- The User define an experiment that needs a certain beam:



Dataset Generation

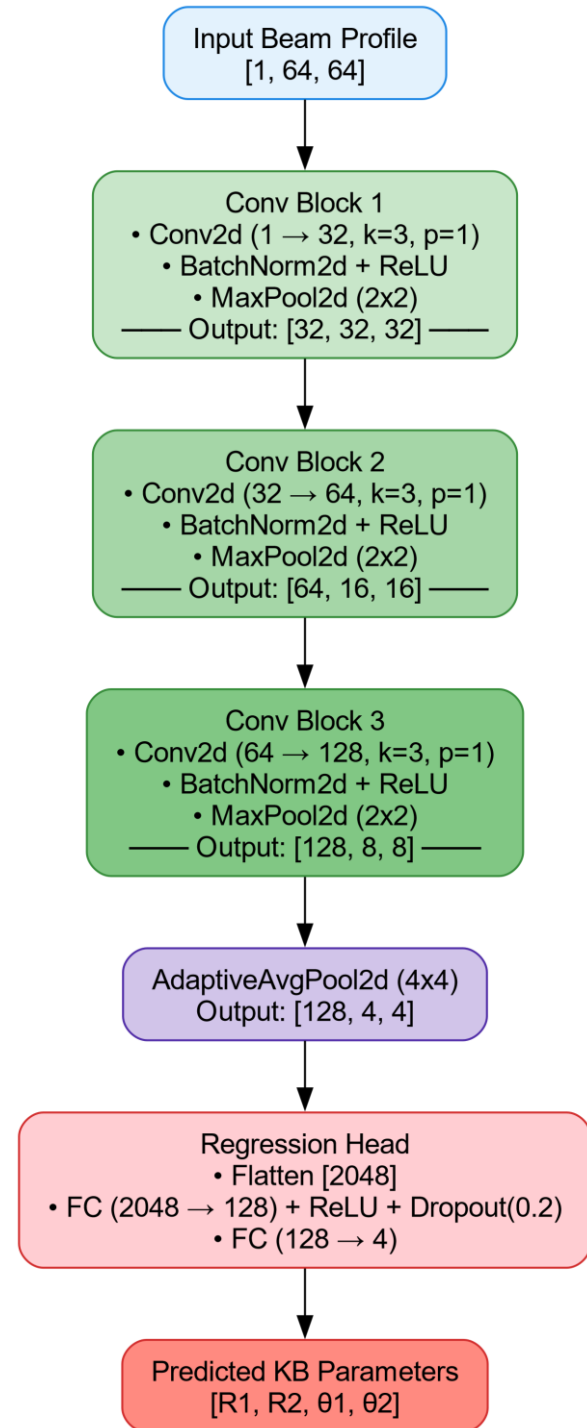
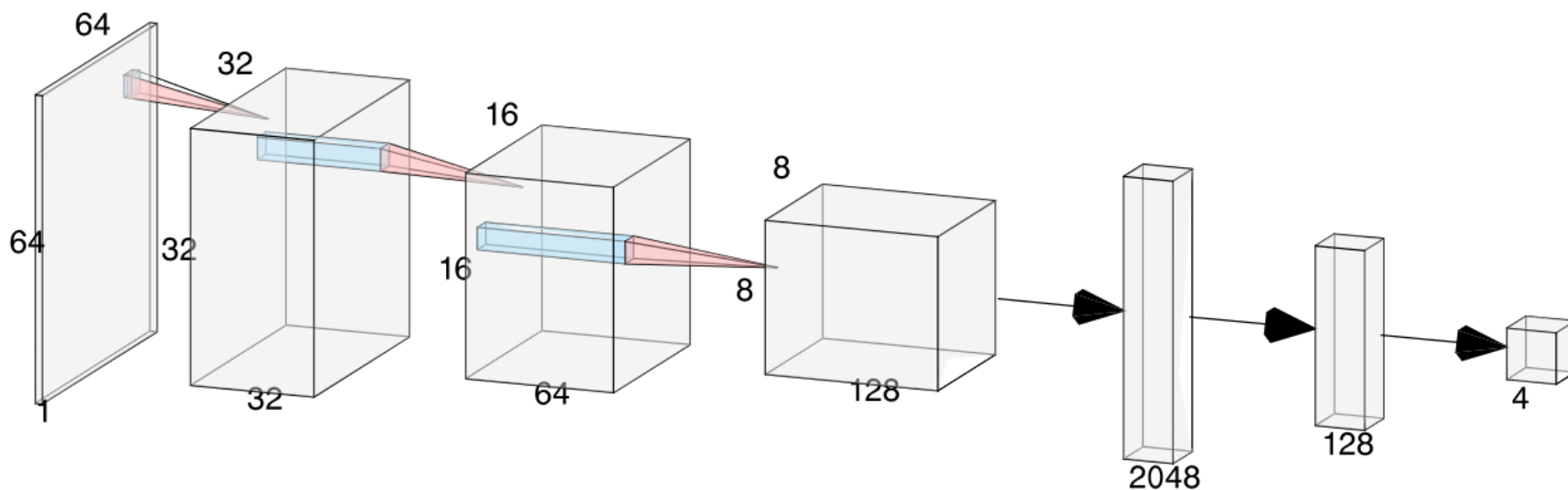
- From **Test_KB.instr** available on McXtrace:
 1. Make Simulation closer to reality:
reduce mirror size and distance to sample
 2. Find Limits i.e. min max of Beam Shape
Via Optimize of McXtrace on dX , dY
 3. Define parameters space from limits
 $(R_1, R_2, \theta_1, \theta_2) = (10, 10, 3, 3) \rightarrow 900$
with discrete steps
 4. Generate Dataset with mxrun
900 images of size $64 \times 64 \sim 10'$ on PC

64x64 Beam Simulation Samples (Train Set) & Corresponding KB Parameters



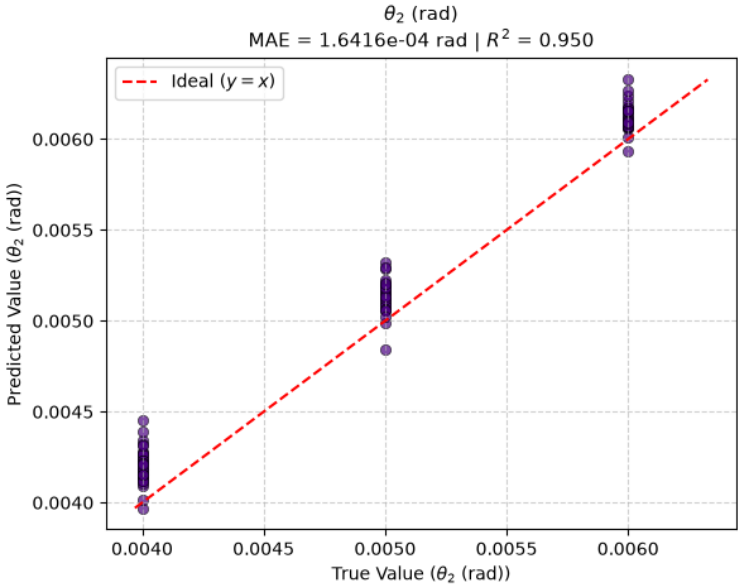
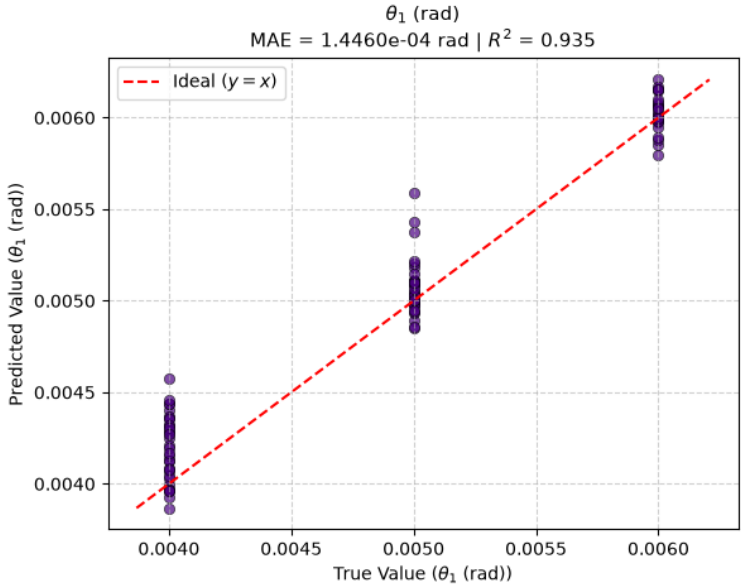
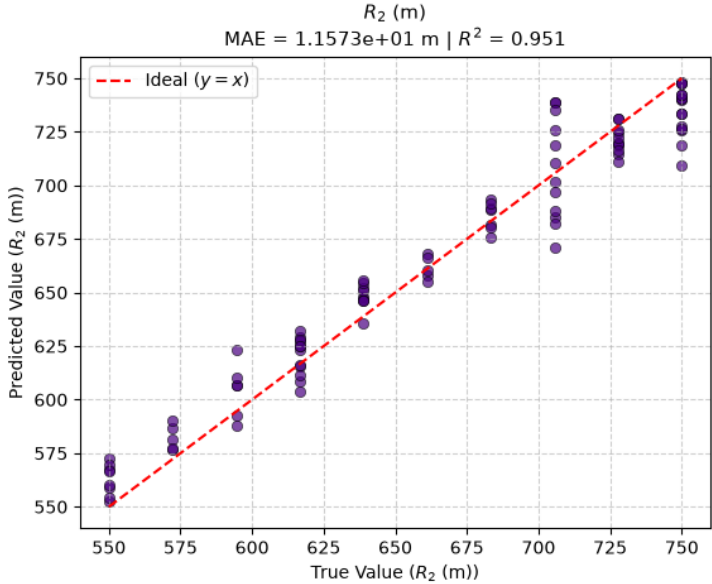
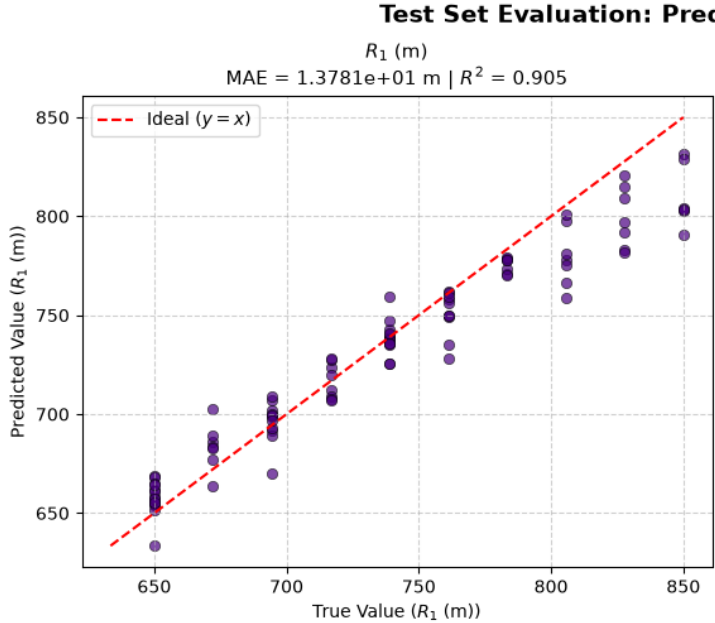
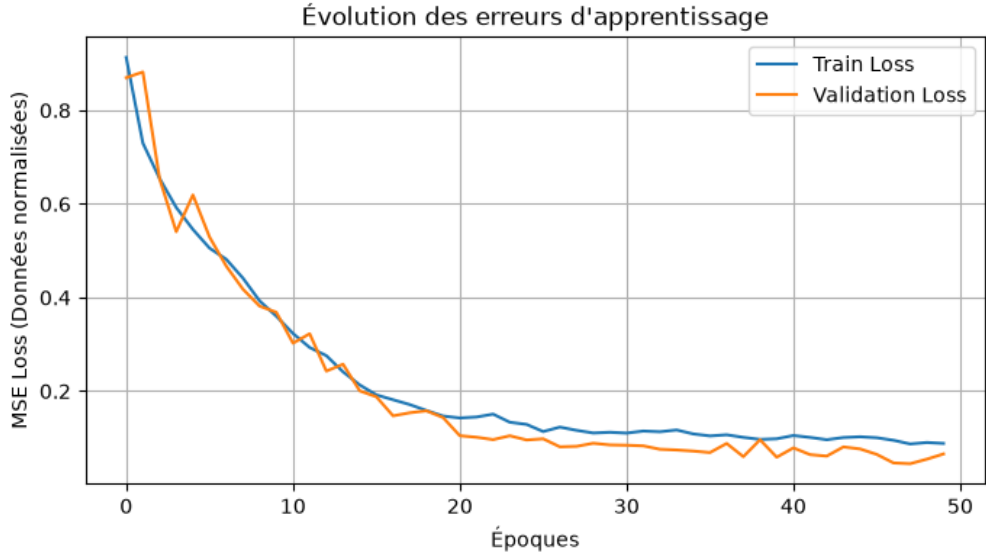
Our NN Architecture

- Convolution Neural Network:



TRAINING 1

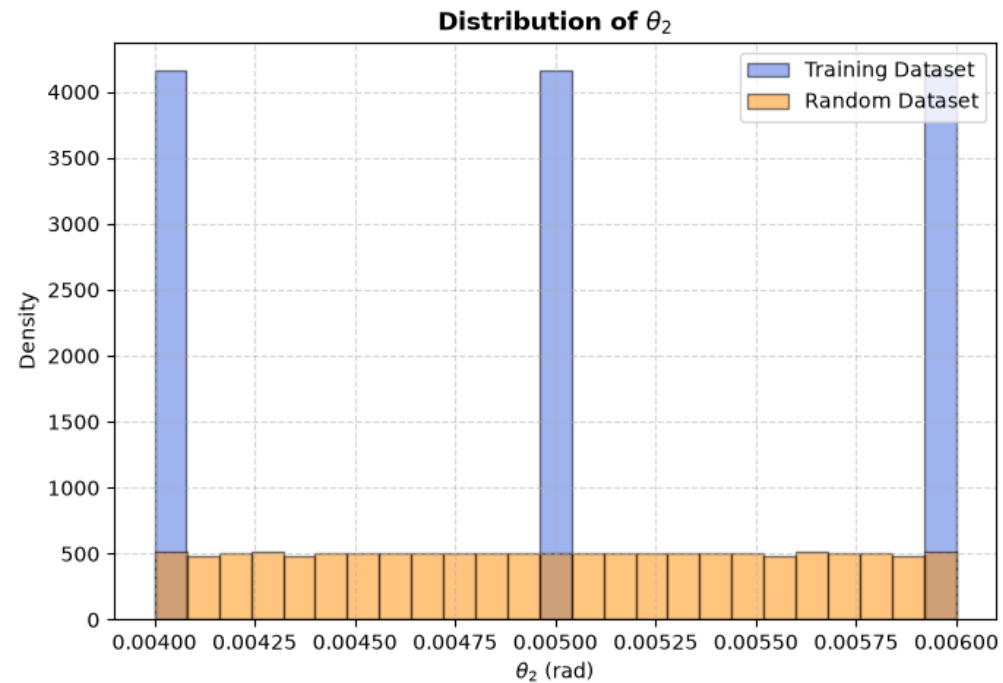
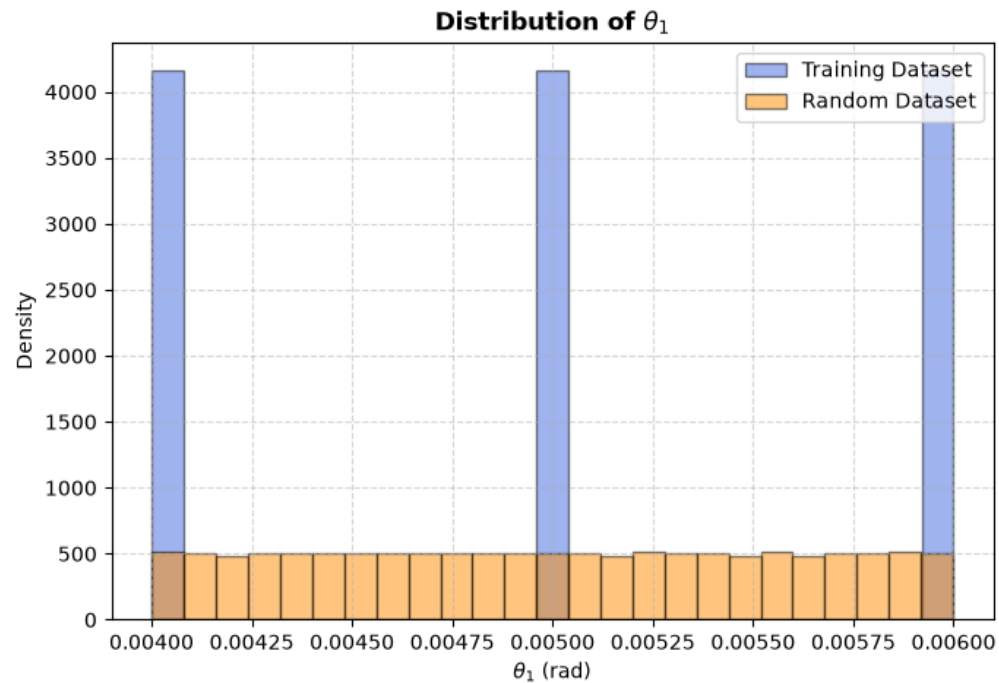
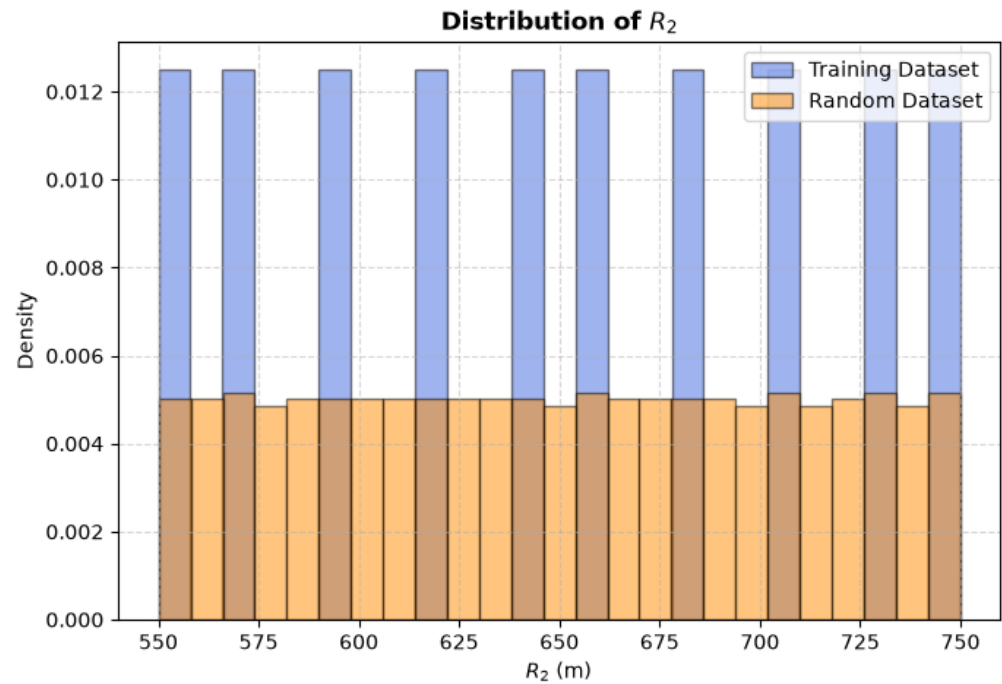
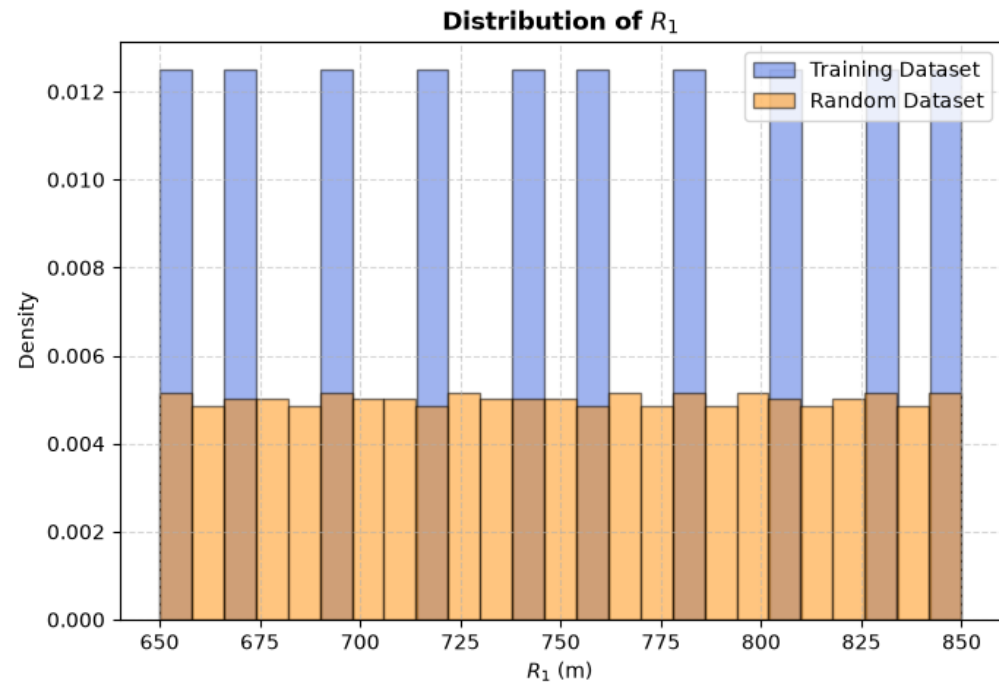
- Trained with **720 images** on **50 epochs** (batch_size=32)



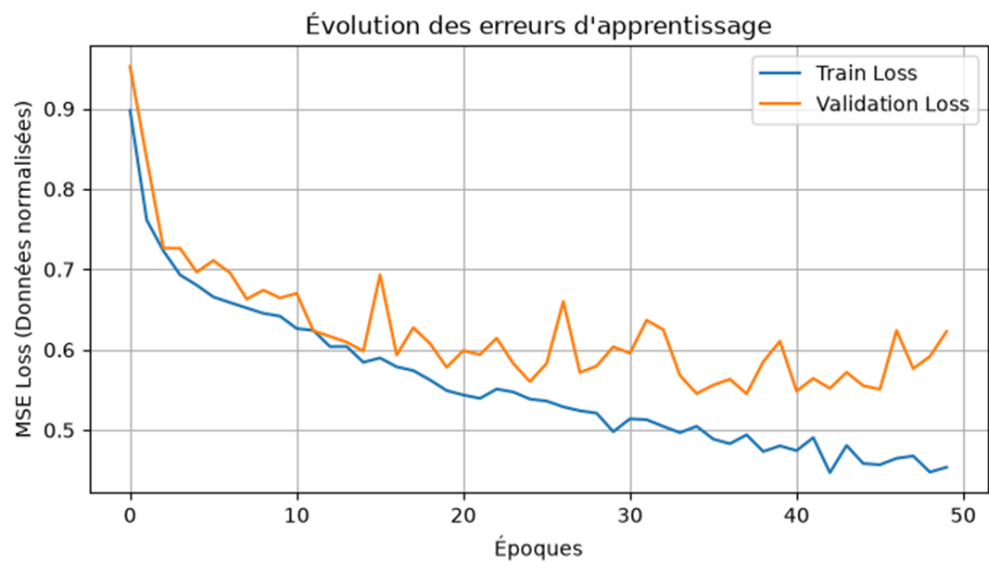
**Finals MAEs
(8.8%)**

- $\rightarrow R_1 = 13.78$ m
- $\rightarrow R_2 = 11.57$ m
- $\rightarrow \theta_1 = 0.14$ mrad
- $\rightarrow \theta_2 = 0.16$ mrad

Parameter Distribution Comparison: Grid vs Random Dataset

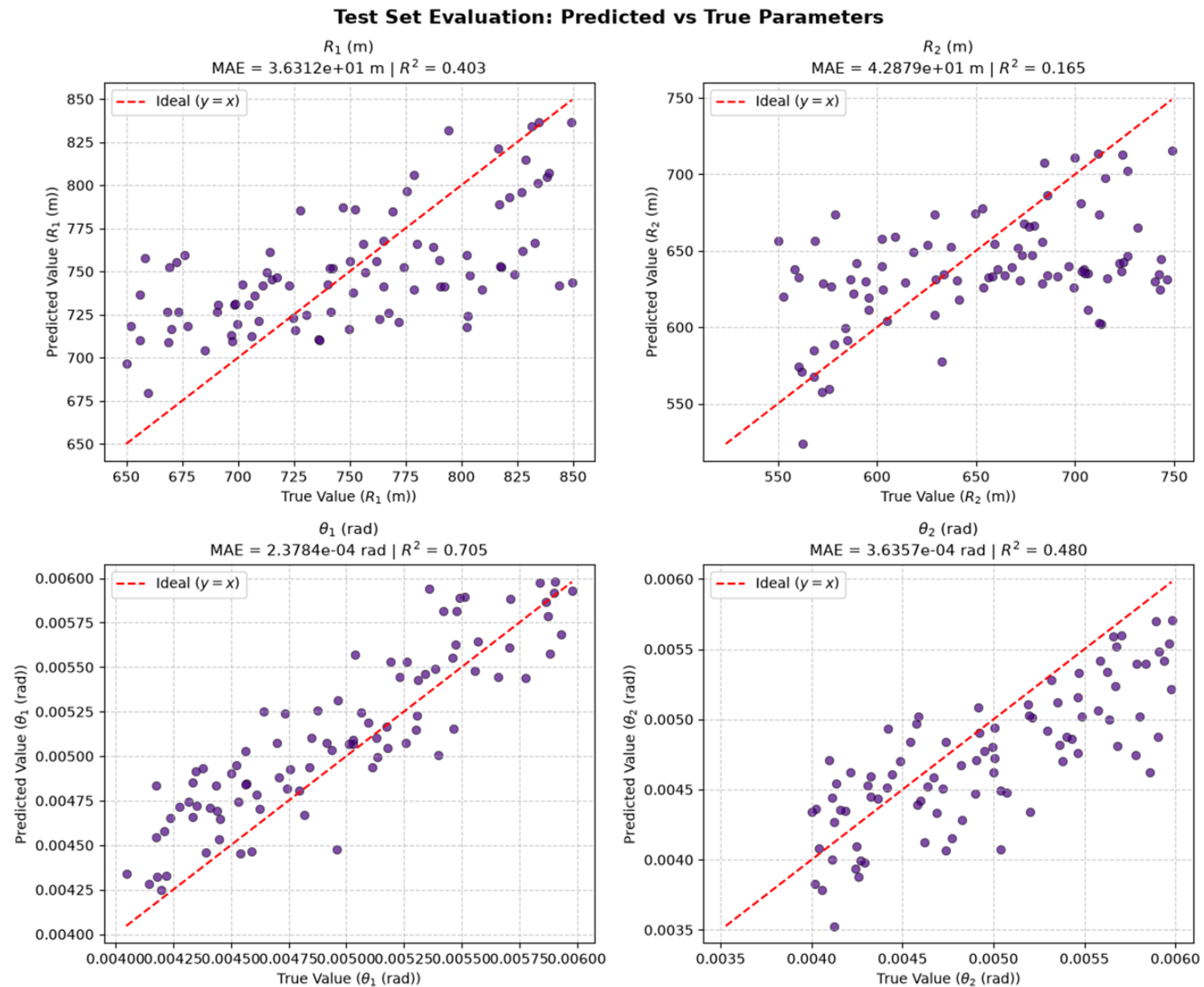


TRAINING 2



Finals MAEs (45% vs 8.8)

- $R_1 = 36.31$ m (vs 13)
- $R_2 = 42.87$ m (vs 11)
- $\theta_1 = 0.24$ mrad (vs 0.14)
- $\theta_2 = 0.36$ mrad (vs 0.16)



- **On the Use Case:**

- Increase Size of the Training Dataset?
- Explore a more complete range of the parameters

- **For the Future:**

- Input more Beamline Optics (DCM, Mirrors, slits) and parameters
- Try to Maximize Intensity for a given Beam Shape
- Extend this method to different Energies